

E-In Son

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RESEARCH INTERESTS

My research focuses on learning-based navigation for robots in real-world environments, spanning traversability estimation in unstructured terrain, visual navigation, and world models. Going forward, I aim to develop world models and navigation policies that generalize across environments and robot embodiments.

EDUCATION

Seoul National University

Mar 2021 – Aug 2027 (Expected)

Ph.D. Student, Dept. of Electrical & Computer Engineering

Seoul, Korea

Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

Konkuk University

Mar 2015 – Feb 2021

B.S., Electrical & Electronics Engineering

Seoul, Korea

Summa Cum Laude (GPA 4.33 / 4.5).

PUBLICATIONS

[1] **Feeling Terrain Before Crossing: World Models for Off-Road Navigation.**

E-In Son, Dong-Wook Kim, Ji-Hoon Hwang, Kangsun Lee, Jisung Bae, Jung-Taak Kim, Seung-Woo Seo.

Under review, 2026. [\[Paper\]](#)

[2] **CanonNav: Disentangling Navigation Behavior from Camera Geometry in Cross-Platform Visual Navigation.**

Dong-Wook Kim, Ji-Hoon Hwang, E-In Son, Mintaek Oh, Seung-Woo Seo.

Under review, 2026. [\[Paper\]](#)

[3] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**

E-In Son^{*}, Jung-Taak Kim^{*}, Seung-Woo Seo.

European Conference on Computer Vision (ECCV), 2026. [\[Paper\]](#) [\[Project\]](#)

[4] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**

Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.

IEEE Robotics and Automation Letters (RA-L), 2025. [\[Paper\]](#) [\[Code\]](#)

[5] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**

Yurim Jeon, E-In Son, Seung-Woo Seo.

IEEE International Conference on Robotics and Automation (ICRA), 2024. [\[Paper\]](#) [\[Code\]](#)

[6] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**

Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.

IEEE Robotics and Automation Letters (RA-L), 2024. [\[Paper\]](#)

[7] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**

Se-Wook Yoo^{*}, E-In Son^{*}, Seung-Woo Seo.

IEEE Robotics and Automation Letters (RA-L), 2024. [\[Paper\]](#) [\[Project\]](#)

AWARDS & HONORS

Earth Rovers Challenge (1st Place) Oct 2024
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
Real-world autonomous navigation competition, AI policies vs. human teleoperators.

Academic Excellence Scholarship 2018 – 2020
Konkuk University

RESEARCH PROJECTS

Challengeable Future Defense Technology R&D Program (Phase 2) 2024 – 2026
Agency for Defense Development (ADD)
Leading the planning stack, from learning-based path prediction to real-time safe replanning, for an end-to-end visual navigation system operating in unknown, hazardous terrain.

Challengeable Future Defense Technology R&D Program (Phase 1) 2022 – 2023
Agency for Defense Development (ADD)
Developed the path tracking module for accurate and stable trajectory following.

TEACHING EXPERIENCE

Seoul National University, Teaching Assistant

- SNU in Silicon Valley Program, joint with UC Berkeley (*Summer 2024*)
- Fundamentals of Random Variables and Probability Processes (*Spring 2023*)
- Network Protocol and Design (*Fall 2021, 2023*)

ACADEMIC SERVICE

Conference Reviewer: IEEE ICRA, IEEE/RSJ IROS, ECCV
Journal Reviewer: IEEE Robotics and Automation Letters (RA-L)

SKILLS

Tools and Languages Python, C/C++, PyTorch, ROS
Communication Korean (native), English

REFERENCE

Advisor: Prof. Seung-Woo Seo sseo@snu.ac.kr [Website](#)